Ablate-and-Measure Breaks When the Measurement Depends on the Ablation

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Abstract

Attribution methods are validated by ablation: remove a component, re-measure model behavior, and check whether the predicted change occurred. This loop assumes the measurement is available whether or not the ablation was applied. On agentic benchmarks that assumption fails. Per-episode metrics are defined only for episodes that reach a scoreable state, and ablations change how often episodes get there. We document this with a 144-episode context-ablation study on τ^2-bench that we could not interpret. The fraction of episodes yielding a defined grounding rate ranges from 50.0% to 91.7% across conditions; for the benchmark's own action-level score it ranges from 4.2% to 33.3%. Because paired testing retains only cells defined in both arms, our *control* condition's measured rate ranges from 0.096 to 0.304 depending on which treatment it is compared against, a 21-point swing in a condition that never changed, exceeding the effect the study was powered to detect. One treatment reverses sign between two defensible analyses of the same episodes. We report this as a failed experiment with a transferable lesson: as data attribution extends to agentic behaviors, the metrics being predicted are exactly the outcome-gated kind that breaks this loop, and four cheap diagnostics reveal whether it is happening.

1 The claim

Contributive attribution asks which data caused a model behavior, and the standard way to check an attribution method is counterfactual: remove the implicated data, retrain or ablate, measure the behavior, and see whether the method predicted the change. Leave-one-out correlation [6], datamodels [5], and the linear datamodeling score [7] all score a method by how well it predicts a model output function f under perturbations of the training set.

The construction has a silent precondition: it presumes f is defined on every perturbed run. For a next-token probability or an accuracy on a fixed test set this is unremarkable, since the metric exists for every input by construction.

It stops being unremarkable when the measured behavior is agentic. A tool-using agent produces a score only if its episode terminates in a state the evaluator will score. Episodes that exhaust a step budget, accumulate tool errors, or overflow a context window yield no measurement at all. They are undefined, not zero. And whether an episode reaches a scoreable state is itself affected by the ablation.

When that happens the two arms are no longer the same population. The analysis conditions on an outcome downstream of the treatment, which is conditioning on a collider [4]. Our contribution is not to name the phenomenon but to show, with measurements, that it is severe enough in agentic evaluation to invert a conclusion, and to propose what to report so that readers can see it.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Attributing Model Behavior at Scale: Data Attribution and Provenance.

We arrived here by failing. We set out to attribute tool-call failures to parts of an agent's context, ran the full grid, and could not interpret the result. We think the failure is more useful than the experiment would have been.

2 What we ran

τ^2-bench [2], retail domain, pinned at the v1.0.1 release commit, which incorporates the benchmark's task-quality corrections. Our attrition is therefore not attributable to the task-specification errors those corrections addressed. Agent and simulated user are both Qwen2.5-7B-Instruct-AWQ under vLLM at temperature 0 with a 200-step cap. Using one model on both sides fixes the user simulator so that differences are attributable to the agent's context. Ablation is an established practice on this benchmark; the original paper reports several.

Six conditions: a full-context control and five ablations removing, respectively, the entire policy document; all policy sections except those governing the task's actions; all tool descriptions; conversation history beyond the last three messages; and policy-framing sentences in the system instruction. Each was implemented by wrapping the episode runner and verified by dumping and diffing the served chat-completion payload for every task. Each condition changed only what it claimed to (Appendix A).

The grid was 12 tasks $\times$ 2 seeds $\times$ 6 conditions, or 144 cells, of which 138 episodes completed. The primary metric was *grounding rate*: the fraction of identifier-valued tool-call arguments appearing in prior context rather than being fabricated. Fabricated identifiers are a documented failure mode on this benchmark family, and the original τ-bench paper reports per-task rates of tool calls carrying non-existent user, product, order, or item IDs [8]. Analysis was pre-registered before the grid ran: paired sign-flip permutation test, $\alpha = 0.05$, Holm correction across five conditions, and a 20-point falsification threshold matched to the grid's minimum detectable effect at 87% power.

3 Definedness is condition-dependent

We chose grounding rate precisely to avoid the benchmark's own scoring gate. That gate is explicit in source: the evaluator returns a null result without reaching action checking whenever an episode does not terminate in one of two clean stop states. Its action-level score consequently exists for between 1 and 8 of 24 episodes per condition, or 4.2% to 33.3% (Appendix D). One condition retains a single defined cell and therefore drops out of every paired contrast without appearing to. A further 21 episodes lost that score to a fault in the assertion judge on our side, again unevenly across conditions, from 1 to 6. That fault was ours, not the benchmark's, and touches only this secondary metric: the grounding rate is computed from transcripts and never invokes the evaluator. A measurement can go missing for reasons unrelated to agent behavior, and an aggregated score does not reveal it.

Grounding rate avoids that gate but not the mechanism behind it. It is undefined when an episode makes no identifier-bearing tool calls, or ends before its grounding pool fills, and that too depends on condition:

Table 1: Episodes yielding a defined grounding rate, by condition.

Condition	Defined (of 24)	Share
Control (Full Context)	22	91.7%
Policy Truncated	22	91.7%
History Truncated	21	87.5%
Schemas Stripped	20	83.3%
System Minimized	13	54.2%
Policy Removed	12	50.0%

Half the policy-removed episodes produce no primary measurement. Choosing a metric outside the benchmark's scoring path reduced the problem without removing it: both metrics require the episode to get far enough to produce the thing being measured.

The driver is visible in termination reasons, where clean terminations range from 4 of 24 to 11 of 24 across conditions, a 29.1-point spread (Appendix B). Our pre-registered rule declared the primary confounded above a 10-point spread. It fired.

Reward-design defects in this benchmark family are documented rather than hypothetical [9], as were task-specification errors in its predecessor's airline domain [3], which prompted the corrections in the release we use. Our point is different and narrower: even a correctly specified reward creates a selection problem if whether it is computed depends on the manipulation. Nor is the practice hypothetical: recent work analyzing false success on this same benchmark discards trajectories with missing rewards as routine preprocessing [1], a step that is standard, reasonable, and exactly the one whose innocuousness we question.

We note that the error-terminated episodes are behavioral, not harness defects. The agent calls real tools with well-formed arguments containing fabricated identifiers until the error budget is exhausted. We verified that tool-call parsing was configured correctly and that every failing call was structurally valid.

4 The control moves

Definedness rates alone do not convey how much this matters. This does.

Our test pairs each treatment cell with the control cell at the same (task, seed), which is standard and reduces task variance. But pairing retains only cells defined in *both* arms, so each treatment induces a different retained subset of the control. The control never changed. Its measured value does:

Table 2: The control condition's own grounded rate, computed on each treatment's retained pairs. Its marginal rate over all 22 defined cells is 0.289.

Paired Against	Pairs	Control's Grounded Rate
Policy Removed	11	0.096
Schemas Stripped	18	0.198
Policy Truncated	20	0.265
System Minimized	11	0.299
History Truncated	19	0.304

Across pairings the control spans 21 points, larger than the 20-point effect the grid was designed to detect, and produced entirely by which cells survived.

The consequence is a sign reversal. For the policy-removed condition, which cuts the served system prompt from 7,056 characters to 357:

- **Marginal:** ungrounded rate 71.7% against the control's 71.1%. Removing the policy document makes grounding *0.6 points worse*.
- **Paired:** grounded rate 0.309 against the control's 0.096 on the 11 shared cells. Removing it makes grounding *21.2 points better* (raw $p = 0.047$, Holm-adjusted $p = 0.235$).

Same episodes, same metric, opposite conclusion. Neither is a coding error. They describe different populations, and nothing in either table tells a reader which population they are looking at.

For completeness: under the pre-registered test no condition is significant after correction, with a minimum adjusted p of 0.235, and none crosses the 20-point threshold under worst-case or best-case imputation of lost episodes (Appendix C). At 87% power for 20 points we can exclude large effects and nothing smaller, and even that bounded null describes the observed subpopulation rather than the intended one.

5 Why this generalizes to data attribution

We ablated context, not training data, and do not claim otherwise. The transferable part is the validation loop, which is shared.

Evaluating a contributive attribution method means predicting how a behavior changes when implicated data is removed, then removing it and checking. Where the behavioral measurement is defined for every input, such as a loss, a probability, or an accuracy on a fixed set, the loop is sound. As attribution moves toward agentic behaviors, which several of this workshop's motivating applications require, such as auditing what data caused an agent to take an action or tracing a tool-use failure to its provenance, the measured quantity becomes an episode-level outcome of exactly the gated kind we document.

At that point the counterfactual check inherits the problem. Removing training data may change how often an agent's episodes terminate scoreably. The retained comparison set then differs between arms, the attribution method's predictions are scored against a shifting population, and the method can be credited or blamed for a selection effect. Nothing about the attribution method is wrong; the harness that validates it measures something other than what it reports.

6 Four things to report

The problem is cheap to detect. We propose that any ablation-based evaluation of agentic behavior report, alongside its headline table:

1. **Per-condition metric definedness.** How many episodes produced a value, not just the mean of those that did. A condition contributing half as many measurements as the control is itself a result.
2. **Per-condition termination distribution.** Clean stops, step exhaustion, error exhaustion, context overflow, and crashes. This is where the selection originates.
3. **Control stability under pairing.** If the analysis is paired, report the control's value on each retained subset. If it moves, the comparison is not what it appears to be.
4. **Bounds rather than exclusions.** Report the primary with dropped episodes imputed at both extremes. If the bounds straddle the decision threshold, say so.

These complement rather than duplicate existing guidance. The Agentic Benchmark Checklist [9] asks whether a reward is *correct*; our diagnostics ask whether it is *defined*, and whether definedness depends on the manipulation under test. None require extra compute: all four derive from records a benchmark run already produces, and we report all four for our own grid.

7 Limitations

One 7B model, one benchmark domain, 12 tasks and 2 seeds. The mechanism is structural and we expect it to recur, but its magnitude here is specific to this setup.

The confound applies to our own null. Section 4's result inherits the problem described in Section 3. We report it as evidence about the design, not about the hypothesis.

Two deviations from our pre-registration, disclosed rather than smoothed over. The rule for scoring error-terminated episodes was written after pilot-level accuracy had been seen; it was derived by symmetry with an existing frozen rule, but it was not written blind. The per-cell aggregation function required by the frozen primary test did not exist when the plan was frozen, and was implemented after marginal point estimates had been seen; the metric, test, α, and correction family were unchanged. The combined attrition statistic that triggered our confound rule was operationalized after the fact, so we publish raw per-condition termination counts for readers who prefer a different definition.

Our truncated-policy condition selects sections using ground-truth actions. It upper-bounds what perfect retrieval would provide and does not simulate a retriever.

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A Ablation verification

Table 3: Served payload per condition, verified on all tasks by diffing against the control. Policy-truncated length varies by task.

Condition	System Prompt (chars)	Tools	Tools w/ Descriptions	History Sent
Full Context	7056	16	16	Full
Policy Removed	357	16	16	Full
Policy Truncated	2311 to 4621	16	16	Full
Schemas Stripped	7056	16	0	Full
History Truncated	7056	16	16	Last 3
System Minimized	6913	16	16	Full

B Termination distribution

Table 4: Termination reason counts over all 144 cells. Crashes account for the 6 episodes that did not complete.

Condition	Normal	Agent Error	Step Cap	Context Overflow	Crash
Full Context	11	9	3	0	1
Policy Removed	11	7	4	0	2
Policy Truncated	11	4	8	0	1
Schemas Stripped	10	11	3	0	0
History Truncated	5	17	2	0	0
System Minimized	4	9	6	3	2

Table 5: Paired sign-flip permutation test on grounded rate, Holm-corrected across five conditions.

Condition	Pairs	Control	Treatment	Diff	Raw p	Holm p
Policy Removed	11	0.096	0.309	-0.212	0.047	0.235
Policy Truncated	20	0.265	0.240	$+0.025$	0.816	1.000
Schemas Stripped	18	0.198	0.261	-0.063	0.485	1.000
History Truncated	19	0.304	0.180	$+0.123$	0.135	0.542
System Minimized	11	0.299	0.208	$+0.090$	0.517	1.000

C Pre-registered primary test

Positive difference means the condition grounded worse than control.

Crash-exclusion bounds, given as change against control under worst-case and best-case imputation: policy removed $+3.4$ and -6.6; policy truncated $+1.4$ and $+1.4$; schemas stripped $+0.8$ and $+5.1$; history truncated $+7.8$ and $+12.2$; system minimized $+10.7$ and -12.7. No condition's bounds cross the 20-point threshold.

D Action-level score definedness

Table 6: Episodes yielding a defined action-level score, out of 24 per condition. "Not Reached" covers episodes whose termination reason prevented action checking. "Evaluation Failed" covers episodes lost to a fault in the assertion judge on our side. Counts derive from the evaluator note stored in each released record.

Condition	Scored	Not Reached	Evaluation Failed
Policy Removed	8	13	3
Policy Truncated	8	13	3
Full Context	7	13	4
Schemas Stripped	4	14	6
System Minimized	3	20	1
History Truncated	1	19	4

E Marginal grounding rates

Ungrounded argument rate, macro-averaged over episodes with a defined rate: full context 71.1% (n=22), policy removed 71.7% (12), policy truncated 72.6% (22), schemas stripped 73.2% (20), system minimized 76.6% (13), history truncated 80.2% (21). The agent fabricates roughly three of every four identifier arguments in every condition, including the control.

F Power

Minimum detectable effect on grounding rate at 12 tasks and 2 seeds, by simulation with 60 replicates per point: 5 points gives 0% power, 10 points 8%, 15 points 43%, 20 points 87%, and 30 points 100%. The 20-point threshold was set to match this curve before the grid ran.

G Reproducibility

Benchmark pinned at the v1.0.1 release commit. Both models are Qwen2.5-7B-Instruct-AWQ under vLLM at temperature 0, with a 200-step cap, on two T4 GPUs, for roughly 16 GPU-hours across two sessions. All 144 episode records, the payload verification dump, and the analysis plan with its frozen and archived sections are available at `https://github.com/abhiprd2000/attrib-context-ablation`.